

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

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Total Cells: 2000 | Model Mode: CLINICAL

1. Introduction

Cellular reprogramming through the use of Yamanaka factors, also known as OSKM (Oct4, Sox2, Klf4, and c-Myc), aims to reset somatic cells to an induced pluripotent stem cell (iPSC) state. The objective of this experiment was to evaluate the effectiveness of repeated OSKM transfections in converting somatic cells into iPSCs under controlled laboratory conditions.

2. Materials and Methods

The experiment involved six consecutive OSKM transfections to induce pluripotency in a population of somatic cells. Despite the planned high-throughput analysis using single-cell variational inference (scVI), no scVI analyses were conducted for this run, focusing solely on the direct outcome of the transfections.

Experimental Interventions:

- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)

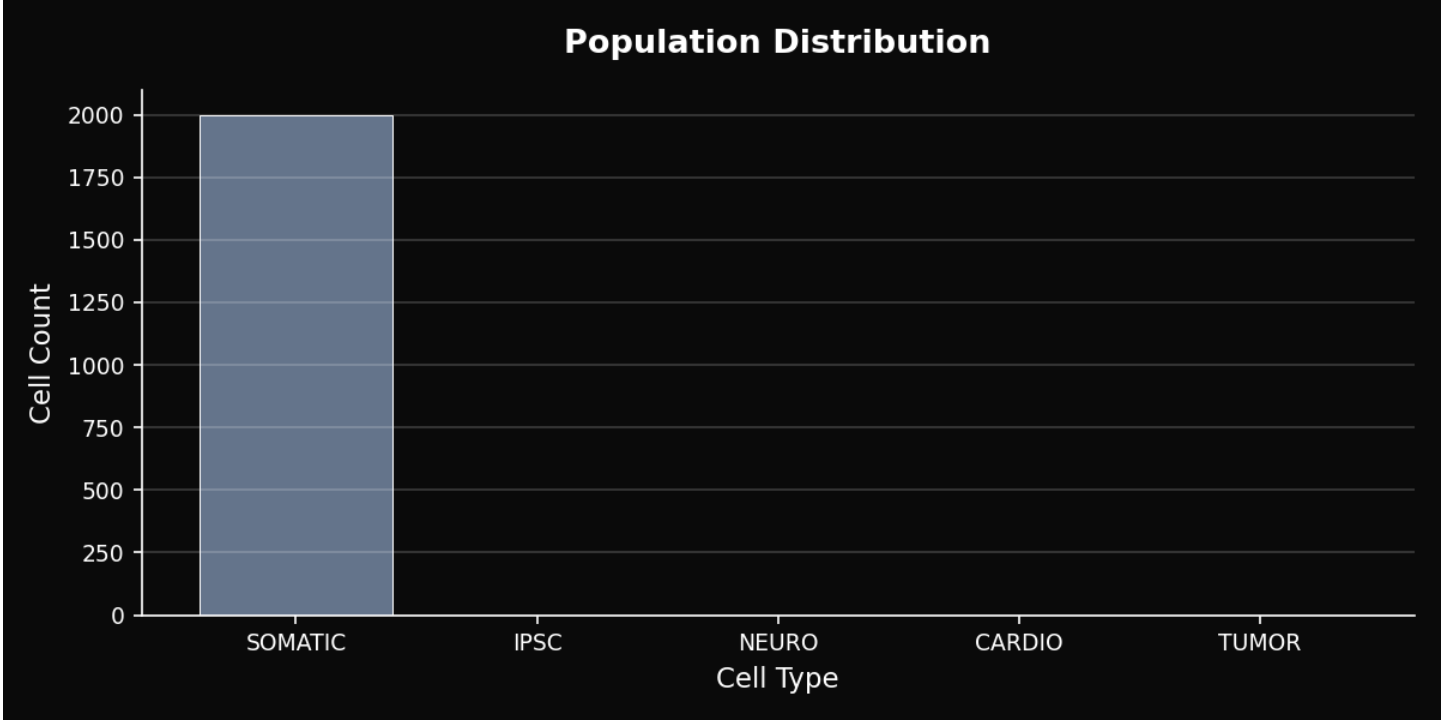
3. Results

The final population analysis revealed a complete absence of induced pluripotent stem cells (iPSCs), as well as other differentiated cell types such as neuro, cardio, or tumor cells. The entire cell population remained somatic, with a total of 2000 somatic cells recorded. This indicates that the OSKM transfections did not achieve the desired cellular reprogramming under the experimental conditions applied.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	2000	100.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	0	0.0%

4. Discussion

These findings highlight a significant challenge in achieving cellular reprogramming using OSKM factors alone in this experimental setup. The lack of iPSC emergence suggests potential limitations in transfection efficacy, cellular resistance to reprogramming, or suboptimal environmental conditions. Further investigation is required to optimize reprogramming protocols, potentially incorporating additional factors or modifying the culture environment to enhance reprogramming efficiency and enable successful iPSC generation.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.